

GENERAL SPECIFICATION G5.41

Diesel Engine Driven Generating Sets

CONTENTS

G5.41 Diesel Engine Driven Generating Sets.....1

 G5.41.1 General.....1

 G5.41.2 Products6

 G5.41.3 Execution.....33

G5.41 DIESEL ENGINE DRIVEN GENERATING SETS

G5.41.1 General

(a) Introduction

- (i) This Section covers the provision of Diesel Engine-driven Generating Sets (DG), including diesel engine, generator distribution switchgear, generator protection and controls, generator synchronizing, load transfer scheme and all ancillary materials and equipment.
- (ii) Perform all work, design supply, deliver and install all materials and equipment and installation methods in accordance with, but not limited to the latest applicable national and international codes, regulations and as listed in General Specification G5.2 Standards, Codes and Regulations.
- (iii) The generator shall include switchgear, generator feeder panel, voltage transformers (VTs), current transformers (CTs), bus bars, protective relays, meters, fast transient and switching overvoltage surge arrestors, Generator controller, engine Instrument and alarm panel (EIAP), high voltage/low voltage switchboards, local control panel for synchronization, load sharing, generator neutral earthing c/w neutral earthing panel and equipment earthing system, and all associated electrical accessories as required for the complete system.
- (iv) The Contractor shall provide a fuel day tank, fuel conveyance pipe, fittings, and isolation valves with stainless steel 304 material.
- (v) The Contractor shall engage an appropriate grade of licenced electrical worker (LEW) to design and supervise all electrical work associated with the DGs. The Contractor's LEW shall endorse all electrical design and construction drawings to be submitted to the S.O..
- (vi) All necessary testing (primary & secondary injection) of protective relays, pressure testing of switchgears & cables, etc. as specified, any other tests which may be required under statutory or professional regulations or that are considered to be consistent with good engineering practices. The tests are to be carried out by the LEW.
- (vii) All cabling works associated with this specification including power, control, instrument and data cables.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

- 1) American Society for Testing and Materials (ASTM): A335, Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service.
- 2) British Standards Institute (BSI)

Standard	Subject
BS 159	High Voltage Busbars and Busbar Connections.

Standard	Subject
BS 476	Fire Tests on Building Materials and Structures.
BS 799	Oil Burning Equipment.
BS 1212	Float Operated Valves.
BS 1363	13A Plugs, Socket-Outlets, Adapters and Connection Units.
BS 2869	Specification for Fuel Oils for Agricultural, Domestic and Industrial Engines and Boilers.
BS 3604-2	Steel pipes and Tubes for Pressure Purposes: Ferritic Alloy Steel with Specified Elevated Temperature Properties.
BS 4999	General Requirements for Rotating Electrical Machines.
BS 5000-3	Specification for Rotating Electrical Machines of Particular Types or for Particular Applications.
BS 5117-2	Testing Corrosion Inhibiting Engine Coolant Concentrate.
BS 5252	Framework for Colour Co-ordination for Building Purposes.
BS 5467	600/1,000V and 1,900/3,300V Armoured Electric Cables Having Thermosetting Insulation.
BS 4999	General Requirements for Rotating Electrical Machines.
BS 6004	PVC-Insulated Cables (Non-Armoured) for Electric Power and Lighting.
BS 7211	Thermosetting Insulation Cables (Non-Armoured) for Electric Power and Low Emission of Smoke and Corrosive Gases when Affected by Fire).
BS 7698	Reciprocating Internal Combustion Engine Driven Alternating Current Generating Sets.
BS EN 10216	Seamless Steel Tubes for Pressure Purposes.
BS EN 10220	Specification for Dimensions and Masses per Unit Length of Welded and Seamless Steel Pipes and Tubes for Pressure Purposes.

Standard	Subject
BS EN 12285	Specification for Fabricated Steel Tanks.
BS EN 60079	Specification for Electrical Apparatus for Explosive Atmospheres.
BS EN 60085	Method for Determining the Thermal Classification of Electrical Insulation.
BS EN 60898	Specification for Miniature and Moulded Case Circuit Breakers.
BS EN 61936-1	Power Installations Exceeding 1 kV
BS ISO 3046-1	Reciprocating Internal Combustion Engines.
BS EN ISO 14120	Safety of Machinery – Guards – General Requirements for Design and Construction of Fixed and Movable Guards.
BS ISO 20816	Mechanical Vibration – Evaluation of Machine Vibration by Measurements on Non-Rotating Parts.

- 3) Diesel Engine Manufacturer's Association (DEMA): Standard Practices for Stationary Diesel and Gas Engines.
- 4) Institute of Electrical and Electronics Engineers (IEEE): 519, Guide for Harmonic Control and Reactive Compensation of Static Power Convertors.
- 5) International Electro-Technical Commission

Standard	Subject
IEC 60034	Rotating Electrical Machines.
IEC 60364	Low Voltage Electrical Installations.
IEC 60479	Effects of Current on Human Beings and Livestock.
IEC 60529	Degrees of Protection Provided by Enclosures (IP Code).
IEC 60590	Determination of the Aromatic Hydrocarbon Content of New Mineral Insulating Oils.
IEC 60896	Stationary Lead-Acid Batteries.
IEC 60947	Low-Voltage Switchgear and Control gear.
IEC 60953	Rules for Steam Turbine Thermal Acceptance Tests.
IEC 61000	Electromagnetic Compatibility (EMC).

Standard	Subject
IEC 61010	Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use.
IEC 61642	Industrial a.c. Networks Affected by Harmonics - Application of Filters and Shunt Capacitors.
IECQ-QC 001005	Register of Firms, Products and Services Approved Under the IECQ System, Including ISO 9000.

6) International Organization for Standardization (ISO)

Standard	Subject
ISO 8528	Reciprocating internal combustion engine driven alternating current generating sets – All Parts.
ISO 9001	Quality management systems -- Requirements

7) SPRING Singapore

Standard	Subject
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.
SS 538	Code of Practice for Maintenance of Electrical Equipment of Electrical Installations.
SS 535	Code of Practice for Installation, Operation, Maintenance, Performance and Construction Requirements of Mains Failure Standby Generating Systems.
SS 532	Code of Practice for the Storage of Flammable Liquids.
SS 555	Code of Practice for protection against lightning.
Electricity Act (Chapter 89A)	Singapore Statutes
NEA	EMISSION STANDARDS FOR OFF-ROAD DIESEL ENGINES

(c) Submittals

- (i) Upon S.O.'s request, the Contractor shall submit to the Superintending Officer (S.O.) for approval 3 copies of a detailed programme showing the schedule for carrying out the following work:
 - 1) Testing of equipment at manufacturer's works;
 - 2) Installation method of statement and risk assessment;
 - 3) Testing and commissioning; and
 - 4) Factory Acceptance Test and Site Acceptance Test reports.
- (ii) Shop Drawings, and Technical Data: As a minimum, submit the following items:
 - 1) Dimensioned outline drawing showing plan and elevations of the proposed Engine Generator Set and drive system, including anchoring requirements.
 - 2) Engine and generator weights.
 - 3) Catalogue information and technical description of the proposed Engine Generator Set. Include proposed materials for the block, heads, valves, rings, cylinders, pistons, crankshaft, and major bearings and wear surfaces.
 - 4) Complete list of accessories provided.
 - 5) Performance curves showing engine efficiency (fuel consumed per kWh output), gross fuel consumption rate, and kW output at design rated output, one-half load, and one-quarter load. Account for design altitude and temperature corrections and for engine parasitic loads.
 - 6) Generator transient and sub transient reactances in per unit.
 - 7) Generator output waveform and Telephone Interference Factor (TIF).
 - 8) Electrical Schematic and Wiring Diagrams:
 - Generator Control Panel;
 - Main Generator;
 - Voltage Regulator;
 - Battery Charging System;
 - Governing System;
 - Generator Distribution Switchgear; and
 - Generator Grounding System.
 - 9) Control Panel instrument identification inscriptions.
 - 10) Engine Generator Set motor starting capability and load acceptance and rejection performance curves.
 - 11) Performance calculations for vibration isolators and confirmation of inertial block sizing prepared by an engineer registered in Singapore.

- (iii) Quality Control Submittals: As a minimum, submit the following items:
 - 1) Operation and Maintenance Manual;
 - 2) Description of parts and service availability;
 - 3) Certification, copies of analyses, or test reports demonstrating appropriate vibration analysis and design in all modes;
 - 4) Certified Factory Test Report;
 - 5) Manufacturer's Certificate of Performance; and
 - 6) Manufacturer's Certificate of Proper Installation.
- (d) Quality Assurance
 - (i) Components shall be designed and manufactured in an ISO 9001 compliant facility.
- (e) Spares
 - (i) Provide recommended common and critical spare parts, materials and special tools as specified in General Specification G2.28 Spare Parts, Tools and Test Equipment.

G5.41.2 Products

- (a) General
 - (i) All engine generators shall be provided and installed in full compliance with all applicable requirements of SS 535.
 - (ii) Generator set ratings shall be in full compliance with ISO 8538-1 and BS 5514.
 - (iii) Generator Voltage Rating: Generator voltage rating shall be 6,600 V or 415 V as shown on the single line diagram.
 - (iv) Generator Set Load Rating: Minimum kW rating as shown in SLD at 0.8 power factor for prime power applications with a minimum load factor of 75 percent. Contractor shall verify the starting capability based on his final design of all the loads.
 - (v) The engine generator set shall be installed in a container or generator room complete with all protections and acoustics suitable of outdoor installations.
 - (vi) The HV and/or LV cables shall be provided. The HV cable connections shall be plug-in type for ease of disconnecting cables during maintenance and / or overhaul when connected at HV level.
 - (vii) All the piping and other mechanical interfaces including fuel pipe shall be able to be disconnected through flexible joints and / or quick couplings during maintenance and / or overhaul.

- (viii) The Generator sets shall be capable of running in parallel, and synchronizing facility from LOCAL and REMOTE in both AUTO and MANUAL shall be provided when more than one DG set is provided. Contractor shall interface correctly with the switchgears to where these generators are connected. Contractor shall study and design synchronizing scheme and propose appropriate solution to SO for approval. ALL DG synchronizing panels shall be interfaced with SCADA and switchgears.
- (ix) The engine and alternator and critical components like governor and AVR shall be of established makes with field proven reliability with local technical support for repair and prompt availability of spare parts. Preference will be given to engine with low specific fuel consumption and low emission. Local technical support for repair and prompt availability of spare parts will also be considered as one of the evaluation criteria.
- (x) The generating unit shall be adequately sized to start all required step loads. Contractor shall verify all steps and submit generator performance characteristics for approval prior to manufacturing.
- (xi) The generator shall further include all other apparatus, appendages, attachments, appliances and connections to make the plant complete in every part and detail to fulfil the intended requirements of the Specifications.
- (b) Manufacturers
 - (i) General
 - 1) Normally engaged in the manufacture of engine generators with minimum of 10 years' experience in the design and manufacture of engine generators.
 - 2) Certified to ISO 9001 The generator set manufacturer shall be certified to ISO 9001 international Quality Standard and shall have third party certification verifying quality assurance in design & development, production, installation, and service, in accordance with ISO 9001.
 - 3) Generators shall be Type Tested design in accordance with IEC standard.
 - 4) DG set shall meet EPA (USA) Tier II and/or in compliance with latest regulation stipulated by NEA (Singapore).
- (c) Service Conditions
 - (i) Altitude: <1000 metres Above Mean Sea Level.
 - (ii) Ambient Temperature at Engine Generator Set: 40 degrees C and 90% Relative Humidity.
 - (iii) Any reference to 'rated kVA' shall mean the 'Prime Rating', that is, the rating when operating under the site conditions prevailing and climatic conditions prevailing in Singapore.
 - (iv) Sufficient inertia shall be provided in the rotating parts of the generating unit so that cyclic irregularity is in accordance with BS ISO 3046.
- (d) Engine Generator Set Vibration Design
 - (i) Meet applicable ISO, SS, IEC, BS, or DEMA vibration standards.

- (ii) Use vibration analytical techniques to determine the shaft critical speeds, and to develop bearing design and shaft balancing to mitigate vibration.
- (iii) Vibration Monitoring System shall be provided in accordance with manufacturer standard and shall be integrated in to the control and monitoring system of the engine generator set.
- (iv) Apply torsional analysis and design to mitigate torsional vibration.
- (v) Vibration limits shall be in accordance with ISO 8528-9, Table C.1, and are stated as root mean square (rms) values:
 - 1) Engine:
 - Vibration Displacement: 0.72 mm; and
 - Vibration Velocity: 45 mm/sec.
 - 2) Generator:
 - Vibration Displacement: 0.32 mm; and
 - Vibration Velocity: 20 mm/sec.
- (e) Engine
 - (i) General
 - 1) Manufacturer's standard design, unless otherwise specified;
 - 2) Engine parts designed with adequate strength for specified duty;
 - 3) Engine shall be 4 cycle, radiator cooled; and
 - 4) Meet requirements of the SS, IEC, BS, or DEMA Standard Practices for Stationary Diesel and Gas Engines.
 - (ii) Emissions: Compliant with EPA (USA) Non-Road Diesel Emission Standards – Tier II and latest NEA (Singapore), Pollution Control Guidelines.
 - (iii) Type and Rating: Diesel Cycle, 4-stroke type and driving a synchronous alternator at a speed of 1,500 rpm. The power rating of the engine at its minimum tolerance level shall be sufficient to drive the alternator and all connected accessories at 110 percent of rated generator set outputs.
 - (iv) The engines shall be closed circuit water-cooled and shall be of the solid injection compression ignition type operating on the four stroke cycle. They shall be pressure charged and intercooled.
 - (v) Each engine shall be capable of delivering the prime rated output of its associated generator including 10 per cent overload for one hour in each twelve hour period. The exhaust temperature at the cylinder outlet shall not exceed 550°C.
 - (vi) Each engine shall be directly coupled to its generator and both shall be mounted on a fabricated baseplate with anti-vibration mountings arranged for installation on structural steel reinforced concrete plinth or foundations. The structural steel reinforced concrete plinth or foundations shall be provided by the Contractor. The structural steel reinforced concrete plinth or foundation shall be designed and endorsed by a Professional Engineer.

- (vii) The flywheel together with the alternator rotor shall be of sufficient moment of inertia to confine the cyclic irregularity and angular deviation to such values to enable the Plant to operate at the duties specified and in any case shall not be outside the limits stated in BS ISO 3046. The engine operating speed shall not be near any of the critical speeds. The engines shall be eminently suitable in all respects for parallel operation of the alternators with each other.
- (viii) Manual flywheel barring facilities shall be provided for maintenance purposes and shall include cylinder head compression release devices. Numbering of each cylinder head shall be indicated and visible from the floor level.
- (ix) Each engine shall be complete with all auxiliary and ancillary equipment to provide a complete, safe, reliable and efficient working installation.
- (x) Starting System
 - 1) Type: Automatic, using a 24-volt battery-driven starter acting in response to the Control Panel.
 - 2) The engine shall be arranged for D.C. electric motor starting. The electric starting equipment shall comprise an electric starter, associated hardware and start repeat relay.
 - 3) The engine shall be arranged for starting from any crankshaft position.
 - 4) 24V D.C. electric starting shall be provided and a suitably rated starter motor shall be fitted to the diesel engine. The electric motor shall be capable of high torques for short periods.
 - 5) When starting, the battery shall be automatically isolated from the charger and then reconnected to allow recharging from the generator power. The Contractor shall submit calculations to show the current drawn when starting the generator and that the battery bank is correctly sized.
 - 6) Batteries:
 - Deep discharge Nickel Cadmium type;
 - 200 amperes hour capacity minimum;
 - Capable of providing cranking current for six full cranking cycles at 10 seconds duration each repeated a minimum of six times;
 - Housed in an acid-resistant frame for mounting on a concrete pad adjacent to the engine. For containerized or trailer-mounted generators, the batteries shall be installed within the container on a suitable steel frame; and
 - Located such that maintenance and inspection of the engine is not hindered.
 - 7) Battery Charger:
 - Float type, with taper charging capability, a float range of 2.17 volts/cell, and an equalisation range of 2.33 volts/cell, minimum;
 - Include ammeter, voltmeter, fused AC input and DC output, pilot light, current failure relay, low/high DC voltage alarm relay, and surge suppressers;

- Alarm relay dry contacts rated 5 amperes at 230 V AC;
- A battery monitoring system shall be provided which initiates alarms when the DC control and starting voltage is less than 24 V DC or more than 32 V DC. During engine starting, the low voltage limit shall be disabled and if DC voltage drops to less than 14.4 V DC for more than 2 seconds a weak battery alarm shall be initiated. Voltage set points for over and under DC voltage shall be adjustable;
- Power Supply: 230 volts AC; and
- Battery charger, battery status and faults shall be communicated to SCADA system.

(xi) Fuel and Governing System

- 1) The engine shall normally be run on Diesel Grade A-1 fuel. The fuel which will be used shall have the following typical characteristics:

• Specific Gravity @ 15oC	Max 0.87
• Cetane Number	Min 48
• Kinematic Viscosity @ 40°C, cst	1.7 to 5.5
• Sulphur % wt	Max 0.01
• Water % vol	Max 0.05
• Sediment % wt	Max 0.01
• Flash Point PM (cc) °C	Min 66
• Copper corrosion 3 hr/100 °C	Max No.1
• Gross calorific value	Min 45,000 kj/kg
- 2) Fuel Pump:
 - Engine-driven;
 - Capable of pumping fuel from above ground day tank to the engine; and
 - Suitable for operation with a 4-metre suction lift.
- 3) The engines shall be closed circuit water-cooled and shall be of the solid injection compression ignition type operating on the four stroke cycle. They shall be pressure charged and intercooled.
- 4) Day Tank (Welded Steel Tank):
 - Day Tank shall be provided individually for each engine generator set. Welded steel tank shall be mounted on the steel structure with adequate foundation and bund wall around the tank for oil containment. The support structure for the day tank shall be designed and installed by the Contractor, subject to approval of S.O.. Access ladder shall be provided as deemed necessary.

- A sump pit shall be provided within the containment area including level electrode to detect water / oil in the sump pit. This level electrode signal shall be connected to SCADA for alarm to operator in control room.
 - Tank shall be complete with manual hand pump.
 - Capacity: 1000 litres.
 - Shall include level controls and monitoring to provide the following signals with necessary multiplier relays as required for the safe and reliable operation of the fuel system:
 - Low-low level alarm (NC, close on alarm) for monitoring of status and alarm in SCADA, and Generator Trip to protect generator from running dry;
 - Low level, dry contact (NO) for fuel oil fill on alarm for monitoring of status in SCADA;
 - High level, dry contact (NC) for fuel oil fill cut-off alarm in SCADA; and
 - High-high level alarm (NO close on alarm) for fuel cut-off alarm in SCADA and auto cut-off of the fuel forwarding pumps.
 - External
 - Grit blast to BS EN ISO 8501 second quality or Swedish Standard 2.5 to a blast profile of 50 - 75 micron;
 - Apply one coat of an inhibited iron oxide pigmented two pack epoxy blast primer to a dry film thickness of 25 micron;
 - Apply one coat of two pack inhibitive primer to a dry film thickness of 75 micron; and
 - Apply two coats of two pack coal tar cured epoxy coating to a total dry film thickness of 300 micron.
 - Internal
 - Grit blast to BS EN ISO 8501 second quality or Swedish Standard 2.5 to a blast profile of 50 – 75 microns;
 - Apply one coat of a two-pack chemical cured holding primer to a dry film thickness of 25 micron; and
 - Apply one coat of a suitable two-pack tank lining to dry film thickness of 125 micron.
- 5) The tank shall be supplied and installed complete with a feed valve and filler cap, shut-off valve, drain plug, vent pipe, access ladder and fuel content gauge. The fuel content gauge shall be calibrated in graduations of 10 litres.
- 6) A fuel strainer shall be incorporated into the fuel transfer pipework immediately before the tank.

- 7) An efficient fuel/water separator with replaceable filter element which shall be approved by the S.O. shall be provided. A water level indicator and a water drain cock shall be incorporated in the fuel/water separator.
- 8) The tank shall be fitted with an intrinsically safe type of fuel level sensing system with low level and high level switches to start/stop the fuel pump automatically. The tank shall also have a manual float switch to shut off the fuel tank at high level. The manual float switch mechanism shall also trigger a high level alarm to alert the station operator. The alarm shall include a wall-mounted siren and a flashing alarm indication lamp on the Local Control Panel. The siren can be silenced by pressing an Acknowledge push button on the Local Control Panel, but alarm lamp shall remain lit until the fuel level in the Daily Fuel Tank has dropped to normal operation level. The fuel level shall be visible through an impact resistant glass tube connected to the tank. A level gauge plate calibrated to an accuracy of $\pm 1\%$ shall be fitted next to the glass tube. The level gauge shall be provided with metallic guards and isolation valves which shut-off upon breakage of level gauge.
- 9) A fuel flow meter shall be fitted from the day service tank to the engine. All fuel oil flow meters shall be of the rotary positive displacement type with an accuracy of better than $\pm 0.5\%$ over the flow range. The counter shall be of straight reading type and the minimum unit shall be 1 litre. Each flow meter shall be provided with a suction strainer, an air eliminator, bypass and isolating ball valves.
- 10) The Contractor shall supply and install all fuel oil pipework and valves to form a complete installation.
- 11) The fuel pipework, fittings, isolating valves, drain valves, y-strainers, non-return valves and pressure gauges shall be constructed of stainless steel 304 materials.
- 12) The Contractor shall ensure that all materials used are compatible with the intended requirements.
- 13) Fuel tank with capacity 400 litres and above shall be registered with SCDF. Contractor shall make the necessary arrangements to engage a qualified person or P.E. to design and submit all the necessary drawings and documents required to register with SCDF. All payments to register the day service tank with SCDF including producing drawings, qualified person charges, registration fee, licence of day tank, licence fee required for the number of years required by SCDF for registration, etc shall be at the Contractor expenses and shall be deemed to be included in the contract sum.
- 14) Contractor shall carry out all interfaces with Fuel Transfer Systems.
- 15) Day Tank Manual Fill System:
 - Furnish and install manual isolation valve, and fuel strainer on day tank;
 - Local fuel oil level indicator shall be provided on each day tanks;

- The manual Local Control Station with LOCAL/OFF/MANUAL Selector Switch, START, STOP and E-STOP push buttons and RUN, STOP, TRIP, POWER ON indicating clustered LED type lamps shall be provided for local controls. The Local Control Station enclosure shall be EX-d type with IP 66 ingress protection;
 - All control functions shall be provided with the generator system and shall be powered from the generator control system battery; and
 - All necessary instrumentation and devices, as recommended by manufacturer, and as required for safe and reliable operation and maintenance shall be provided.
- 16) Provide fuel oil cooler between engine and day tank to maintain return fuel temperature to below 68 degrees C.
- 17) Fuel Oil Filter: Replaceable element type.
- 18) Fuel Connections to Engine: Flexible hose, suitable for application.
- (xii) Governing System
- 1) Governor:
- Electronic engine management system with digital electronic governor system;
 - Regulates speed as required to hold generating frequency within tolerable limits and within 5 percent of nominal design speed; and
 - Accessories:
 - Manual speed control device; and
 - Positive over speed trip switch.
- 2) A digital electronic governor system shall provide automatic isochronous frequency regulation. The governing system dynamic capabilities shall be controlled as a function of engine coolant temperature to provide fast, stable operation at varying engine temperature conditions. The control system shall actively control the fuel rate and excitation as appropriate to the state of line generator set. Fuel rate shall be regulated as a function of starting, accelerating to start disconnect speed, accelerating to rated speed and operate in various isochronous or parallel status.
- 3) The governor shall be capable of performing across a wide speed range and offer fast engine response to speed or load change and stable operation.
- 4) Separately programmable proportional, integral, and derivative gains shall be provided for tailoring governor response to various engine applications with LCD or LED display to indicate selected control parameters and parameters values. It shall have an internal FAILSAFE that instantly reacts to DC power interruption and loss of speed reference signal. Under any of the events, the actuator shall return to minimum fuel position.
- 5) The engine operating speed shall not be near any of the critical speeds. An independent over-speed trip device shall be provided for each engine.

(xiii) Jacket Water Cooling System

- 1) Engine-mounted or remote mounted radiator with jacket water pump, fan assembly, fan guard, and duct flange outlet.
- 2) Engine thermostat to regulate engine water temperature as recommended by the manufacturer. All power supply and control functions required by a remote mounted radiator shall be included as an integral part of the generator package.
- 3) The cooling system for the generating set shall be cooled by means of a set mounted radiator complete with a reverse flow fan. The skid mounted radiator and cooling system shall be rated for full load operation in 40 degrees C ambient as measured at the generator air inlet, based on 500 Pa external static head.
- 4) Remote radiator shall be sized based on a core temperature which is 40 degrees C higher than the rated operation temperature. Radiator shall be provided with a duct adapter flange.
- 5) The circulating pumps shall be of the centrifugal type and shall be engine mounted and engine driven. The jacket water system shall incorporate a thermostat and a thermostatic diversion valve.
- 6) The radiator shall be fan cooled and thermostatically controlled. The fan shall be suitably rated, and shall be engine driven. An air duct shall be provided to discharge to the outside the air after cooling the radiator. The duct shall be connected to the radiator via a flexible joint.
- 7) The cooling system shall use a 50/50 ethylene glycol/water mixture. Fan cowl and protection guards for rotating parts shall be guarded against accidental contact.
- 8) Provisions shall be made to enable the radiator water to be drained out.
- 9) The radiator fill cap, the cooling system drain cocks and the fan belt tension adjustment shall be easily accessible for maintenance.
- 10) The engine cooling system shall be designed to maintain constant jacket water temperatures before and after the radiator as well as constant lubrication oil temperatures before and after the lubricating oil cooler. These temperatures shall be within the limits recommended by the engine manufacturer, and shall also comply with the specification herein.

(xiv) Lubrication System

- 1) Each engine shall have a forced feed lubricating oil system operating on wet sump principle and shall comprise lubricating oil circulating pump, reservoir tank, a coarse suction strainer and discharging through a heat exchanger, filters and thermostatic by pass valves.
- 2) The filters shall be of duplex dual flow type. They shall have easily removable type elements and a changeover lever to enable changing of filter elements without shutting down the engine. Filters grade shall not be more than 75 microns or better type centrifugal filters may be provided.

- 3) The pressure pump shall be of the gear type and shall be engine driven. The pump shall be protected from excess pressure by a pressure relief valve built into the pump housing or by other suitable means.
- 4) The system shall be complete with pipework, fitting, valves and filters. Automatic periodic oil circulation once every 1 to 3 hours for 10 to 15 minutes shall be provided when the engines are not running if recommended by the engine manufacturer.
- 5) Level gauges or other approved means shall be provided on each engine to ascertain whether the volume of oil in an engine is satisfactory even while the engine is running.
- 6) All necessary thermometers, pressure gauges including pressure indication across the filters shall be fitted to the system.
- 7) All high points in the pipework shall be provided with plugged air release bosses and all low points with plugged drain bosses.
- 8) All drains, protection devices, valves and fittings necessary for the safe and efficient operation of the system shall be provided. A drain valve shall be provided at the bottom of the lubricating oil sump to enable the draining of the lubrication oil from the sump.
- 9) A semi rotary hand pump shall be supplied complete with flexible hoses of sufficient length to outside the container, so that oil from the sump drainage tank may be returned to 200 litre drums for storage before disposal.
- 10) The first filling and first charging of lubricating oil shall be included under the Contract and the Contractor shall give his recommendations in regard to any special grades of oil that may be considered necessary. The first oil change shall be carried out by the Contractor.
- 11) The Contractor shall supply all the pipework, fittings, drip tray and valves necessary to provide an efficient and workable system.
- 12) Type: Full-pressure.
- 13) Accessories:
 - Pressure switch to initiate shutdown on low oil pressure;
 - Oil filter with replaceable element;
 - Bayonet type oil level stick; and
 - Valved oil drain extension.
- 14) Oil Cooling System: Water-cooled heat exchanger utilising jacket water.

(xv) Air Intake System

- 1) Each engine shall be provided with a dry type air cleaners with filter service indicator for combustion air supply. Air inlet apertures are to be covered with fine mesh screen panels, which can be easily removed, for cleaning. The differential pressure across the filters shall be indicated. The installation shall be complete with the necessary air ducting between the filter unit and the turbo charger inlet. Ducting shall be complete with means to take up expansion or contraction due to temperature changes and shall be complete with all necessary supports.
- 2) Temperatures of exhaust at entry to the turbo charger shall be indicated with pyrometers at the exhaust branches. Intercoolers shall have air "in" and "out" and water "in" and "out" temperatures indicated. Air pressure before and after the inter cooler shall also be indicated. A water flow indicator shall be provided for each intercooler.

(xvi) Exhaust System

- 1) The engine exhaust shall be ducted to external part of the generating set enclosure. The exhaust duct shall be as direct as possible so as to minimise restriction to gas flow. The exhaust pipe shall be connected to engine manifold by flanged stainless steel bellow unit. The material of the exhaust piece after the stainless steel bellow unit and outside the container or generator room shall be stainless steel 304 material. Residential grade silencer shall be used and shall be mounted as close as possible to the engine. Asbestos-free lagging material shall be used to reduce radiator heat and noise so as to allow operation personnel to inspect or take operation data in close range without danger. All lagging material shall be covered with 0.5mm aluminium jacket, secured with stainless steel screw. The end of the exhaust pipe shall be in the form of a tailpiece to prevent the entry of rain water.
- 2) Individual pyrometers shall be installed at each cylinder exhaust and shall be connected to a multipoint digital type thermometer. They shall be arranged for reading from floor level. The exhaust duct shall be complete with all necessary support structures, rollers and brackets. The exhaust ducting shall be fully lagged and encased in aluminium cladding. The design of the cladding shall be such that the surface temperature shall not be more than 55 degrees Centigrade. The lagging shall be of the Rockwool type.
- 3) The silencer shall be of the reactive type and shall be manufactured from high quality stainless steel. It shall be of rigid design. Suitable supporting cradles shall also be supplied together with the necessary holding down bolts. The exhaust outlet shall be above the roof level of the generator container or generator room. All supporting frames shall be of stainless steel.

- 4) The installation and operation of the generator shall also comply with the Environment Pollution Control (Boundary Noise Limits for Factory Premises) noise regulations and standards as stipulated by the National Environment Agency (NEA). The sound attenuator shall be installed on the discharge side of the radiator. The attenuator shall be constructed with high quality hot dipped galvanized mild steel casing with end flanged. Acoustic splitters shall be aerodynamically shaped, having a bull nose at the inlet and follows by a parallel sided section of varying length and terminated by an optimized tapered section. The splitter and side wall linings shall be of resin bonded fibreglass with hot dip galvanized perforated metal applied over the acoustic lining material. Flanges shall be drilled and tapped to match the flanges of the connecting ductwork. The sound pressure level from the engine shall not exceed 105 dB(A) when measured 1m from the DG. Full details of the noise and vibration reduction accessories shall be submitted to S.O. for approval. The Contractor shall purchase the sound level meter to carry out the above noise measurement. The sound meter shall be handed over to the S.O. after the commissioning of the generator.
 - 5) Silencer: Silencer shall be rated for critical grade silencing.
 - 6) Exhaust Pipe: ASTM A335, Grade P11, or equivalent IEC or BS, standard wall, with fittings selected to match piping materials.
 - 7) Pipe Connections: Welded.
 - 8) Engine Connection:
 - Flanged, flexible, corrugated, stainless steel expansion fitting, specifically suited for diesel exhaust service; and
 - Length as required for flexibility and expansion in the piping arrangement shown.
 - 9) Overall Sound Treatment: The generator shall be housed in an acoustic container or in generator room with acoustic wall and the noise level shall not exceed 85 dB(A) at a circumferential distance of 1m from the container or generator room when the diesel generating set is in operation under full load condition.
- (xvii) Engine Instrument and Alarm Panel
- 1) An Engine Instrument and Alarm Panel (EIAP) of the floor mounting type shall be provided adjacent to respective diesel engine to house all protection and shut down devices, instruments, and status and alarm indicators for the diesel engine. All the necessary piping, wiring and terminal boards shall be mounted in a neat and readily accessible manner with the mechanical and electrical sections segregated.
 - 2) The following instruments and meters shall be installed on the panel, as a minimum if applicable:
 - Digital Tachometer for engine speed indication;
 - Pressure gauge for jacket water (engine inlet);
 - Pressure gauge for cooling water;

- Pressure gauge for lubricating oil (engine inlet);
 - Pressure gauge for fuel oil (engine inlet);
 - Pressure gauge for air receiver;
 - Pressure gauge for charge air;
 - Pressure gauge for intake air (left hand side);
 - Pressure gauge for intake air (right hand side);
 - Thermometer for jacket water (engine outlet);
 - Thermometer for lubricating oil (engine inlet);
 - Thermometer for intake air (engine inlet);
 - Digital type thermometer for the indication of temperature of exhaust gas of cylinders, turbo charger inlet and outlet;
 - A selector switch shall be provided to read any of the above temperatures; and
 - Hour run meter.
- (xviii) All manual pressure gauges and thermometers shall be of a minimum 150mm diameter. The normal operating range shall be marked on every indication meter or gauge. Digital gauge and thermometers is only allowed subjected to the approval by the S.O.. Means of indicating the normal operating range shall be provided.
- (xix) The following control switches/pushbuttons shall also be provided in EIAP.
- 1) Key operated Local/Off/Remote change over control switch to enable control either from EIAP or from Local Control Panel (LCP);
 - 2) Engine start/stop spring return to neutral control switch; and
 - 3) Emergency stop pushbutton (key release type).
- (xx) Isolator switch and Heater On/Off indication lights for stator and exciter heaters.
- (f) Alternator
- (i) General
- 1) Meet requirements of applicable SS or IEC standards;
 - 2) Synchronous salient-pole type with amortisseur winding in rotor;
 - 3) Stator Windings; and
 - Skewed for smooth voltage waveform;
 - Reconnectable, 12 leads; and
 - 105 degrees C winding temperature rise at 40 degrees C ambient.

- 4) A set of anti-condensation heaters shall also be fitted into the stator frame and shall be connected to the 230V supply of the generator battery charger. The heaters shall maintain a temperature inside the machines of about 5°C above ambient during shut-down periods. The control switches and associated lights shall be mounted on the control and alarm panel. The heaters shall be automatically switched off when the engine is running and switched on when the engine stops.
- (ii) Ratings
- 1) Alternator Load Rating: Equivalent to generator set rating;
 - 2) Voltage: As per Single line diagram, frequency 50-Hz;
 - 3) Over speed Capability: 150 percent;
 - 4) Waveform Deviation from Sine Wave: 5 percent maximum;
 - 5) Telephone Interference Factor: 50 maximum; and
 - 6) Total Harmonic Factor: 3 percent maximum.
- (iii) Insulation System
- 1) Class H insulation with Class F rise in accordance with IEC;
 - 2) Vacuum impregnated, epoxy varnish; and
 - 3) Fungus resistant.
- (iv) Excitation System
- 1) Excitation of the field for each generator shall be provided by brushless excitation with direct driven AC main and permanent magnet pilot exciters with controlled static amplifications of the permanent magnet pilot exciters output, applied to the field of the main AC exciter, whose output is rectified by rotating diodes and applied to the generator field. The rotating rectifiers shall be able to withstand surge voltage of more than 3 times normal voltage without damage. The rectifiers shall also be protected by Silicon Carbide Varistor.
 - 2) The excitation system offered shall be of a well proven design. Evidence of a satisfactory operating record of service with machines of larger size and construction to those offered shall be submitted. The system offered shall be based upon a system which will give the highest reliability and performance when operating at the duties specified.
 - 3) Rotating main and pilot exciters shall be of the enclosed type directly driven from the generator shaft. Removable covers, with inspection windows shall be provided such that easy access to connections and rotating diode assemblies is provided.
 - 4) Separate circuits shall be provided to facilitate automatic or manual control of excitation.

- 5) Provision shall be made for automatic follow-up of the manual set point controller to ensure smooth changeover from automatic to manual control. The follow-up equipment shall operate after a short time delay so that if the automatic voltage regulator trips automatically, the excitation control shall set the field current to the value existing prior to the trip, provided that this current is consistent with stable operation of the generator under manual excitation control at the prevailing load conditions.
- 6) The generator and excitation systems, along with all associated equipment, shall be able to sustain rated ceiling excitation, with a bolted three-phase short-circuit on the 6.6kV terminals (for HV generators) or 415 V terminals (for LV generators), for at least 3 seconds without damage. This is to ensure sufficient current flow for satisfactory operation of overcurrent protection and to permit continued AVR operation during, and immediately after system faults. Excitation System: Brushless, PMG (Permanent Magnet Generator type).

(v) Voltage Regulation

- 1) Solid state, three-phase sensing type;
- 2) Adjustable output voltage level to plus or minus 5 percent;
- 3) Conformal coating environmental protection; and
- 4) Accessories: Harmonic filters, or other provisions necessary, for proper voltage regulation when one or more adjustable frequency drives are a part of the generator load.

(vi) Voltage and Frequency Regulation Performance

- 1) The generator set shall include an automatic digital voltage regulation system that is matched and prototype tested with the governing system provided. It shall be immune from miss-operation due to load induced voltage waveform distortion and provide a pulse width modulated output to the alternator exciter. The voltage regulation system shall be equipped with three-phase rms sensing and shall control build-up of AC generator voltage to provide a linear rise and limit overshoot. The system shall include a torque-matching characteristic, which shall include adjustment for gain, damping, and frequency roll off. Adjustment shall be broad angle and made via digital raise–lower switches with an alphanumeric LED readout to indicate setting level. Rotary potentiometers for system adjustment are not acceptable.
- 2) Steady State Voltage Regulation: Less than plus or minus 1 percent from no load to prime rating point.
- 3) Voltage regulation for any constant load shall be plus or minus 0.5 percent. Random voltage variation with any steady load from no load to full load shall not exceed plus or minus 0.5 percent.
- 4) Random frequency variation with any steady load to full load shall not exceed plus or minus 0.25 percent.

- 5) Transient Voltage Dip:
 - Less than 20 percent at sudden application of rated load; and
 - Recovery to rated voltage and frequency within 2 seconds following initial load application.
- 6) Transient Voltage Rise:
 - Less than 15 percent at sudden drop off application of rated load; and
 - Recovery to rated voltage and frequency within 2 seconds following load drop off.
- 7) Steady State Frequency Regulation: Plus or minus 1.5-Hz overload range.
- 8) The AVR shall include the following:
 - Provision to control the generator voltage within the range specified.
 - A balancing circuit with a null meter to permit smooth changeover by an operator from manual to AVR control or vice versa;
 - Parallel running compensation and connections to ensure stable parallel operation and correct sharing of active and reactive power with other generators; and
 - Facilities to modify the AVR droop characteristics such that the change in kVAr loading, resulting from system voltage changes, can be satisfactorily adjusted
- 9) The AVR shall include the following protective devices which shall revert the running AVR to the manual control and also give appropriate alarms:
 - Over excitation protection;
 - Under excitation protection;
 - Over voltage protection; and
 - Over current protection.
- 10) V.T. fuse failure.
- (vii) Motor Starting Capability: 120% full load (minimum) percent generator rating at 0.25 power factor.
- (viii) Variable Frequency Drive Load Capabilities
 - 1) Variable Frequency Drive Steady State Load: 60% full load (minimum) of generator rating;
 - 2) Drive Type: Pulse width modulated inverter; and
 - 3) Total Harmonic Distortion at Adjustable Drive Bus: Limited by drive to 5 percent in accordance with IEC or IEEE 519-81.
- (ix) Short-Circuit Capabilities: Sustain 300 percent of rated current for 10 seconds for an external three-phase bolted fault without exceeding rated temperatures.
- (x) Enclosure: IEC 60529 IP 23 (minimum).

- (g) Instrument Transformers (CT'S & PT'S)
 - (i) Provide Current Transformers (CT) and Potential Transformers (PT) as necessary for the complete system;
 - (ii) Protection CT's shall be 5P20 or Class X for differential protection, and accuracy class 0.5 for Metering. Minimum CT burden shall be 15 VA; and
 - (iii) Protection PT's shall be Class 3P and accuracy class 1.0 for Metering.
- (h) Generator Protection
 - (i) Following protections shall be provided through multifunction numerical Generator Protection relay:
 - 1) 51V VOLTAGE RESTRAINT OVER CURRENT;
 - 2) 51 OVER CURRENT;
 - 3) 87 STATOR DIFFERENTIAL PHASE AND EARTH FAULT;
 - 4) 27 UNDER VOLTAGE;
 - 5) 59 OVER VOLTAGE;
 - 6) 81UF UNDER FREQ;
 - 7) 81OF OVER FREQ;
 - 8) 32 REVERSE POWER;
 - 9) 49 STATOR WINDING TEMP HIGH;
 - 10) 51G EARTH FAULT;
 - 11) 49 WINDING TEMPERATURE;
 - 12) 49 BEARING TEMP HIGH; and
 - 13) 86 GENERAL LOCK OUT.
 - (i) Energy Meter
 - 1) The Energy Meter shall be of microprocessor-based type and with high accuracy over a wide dynamic measuring range. It shall provide harmonic analysis and waveform recording facilities, as well as instantaneous and integrated power system parameters. The device shall be suitable for panel mounting:
 - Following features should be available as a minimum requirement:
 - True RMS measurements suitable for single, two or three phase power systems application;
 - Four quadrant power measurement;
 - Accurate monitoring of instantaneous and integrated values: ± 0.5 percent for current and voltage measurements, ± 0.5 percent for power and demand measurements;
 - Energy metering to IEC 60687 Class 0.5S;

- Maximum and average demand for I a, I b, I c, W, Var, VA, PF, Wh and Varh together with power control information;
 - Maximum and average demand at 5, 10, 15 or 30 minutes intervals;
 - All demand and energy values should be stored in non-volatile memory including maximum demand current and power;
 - Real time harmonic and percentage THD analysis up to the 50th harmonic for each directly measured current and voltage;
 - Waveform recording at least up to 72 cycles for all 6 directly measured currents and voltages, opto-inputs and alarm status; and
 - Waveform capture providing a 2 cycle record of 192 samples/cycle on the selected phase to have harmonic analysis up to the 60th.
 - Large clear LCD display for direct reading of three phase measurements;
 - Display of measured values in primary or secondary quantities;
 - Four alarm/pulse relays;
 - Two status inputs;
 - DC input (suitable for transducer input);
 - Four analogue outputs;
 - Thirty-two alarm settings, high/low and variable trip/reset times;
 - One hundred historical alarm records; and
 - An easy to use PC setting software.
- 2) Communications System Interface: Each protection relay and energy meter shall be capable of communicating to the PMS over an Ethernet connection by the following method:
- Media Connection:
Multi-drop optical fibre connection from relays and power analysis and measurement centre. Each relay shall have 2 integral fibre optic ports for direct Ethernet communication.
- 3) Protocol: TCP/IP, Modbus, or other protocol to suit PMS.
- 4) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational.
- 5) Data Registers: As a minimum, provide the following information as data which can be queried by the Modbus master:
- All measured and calculated parameters that are read at local displays;
 - Protective functions with trip and/or alarm outputs;
 - All control functions required for remote operation of the circuit breakers and bus bar disconnectors;
 - Wave form recording and capture; and
 - Event logs.

- 6) Provide front panel accessible USB port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - View actual values;
 - View unit status;
 - View/edit settings (online and offline);
 - View event record (for troubleshooting);
 - Help screens and step-by-step tutorial; and
 - Provision for installation of firmware upgrade(s) to unit.
 - 7) Provide a completely functioning protection and control system making full use of the features of the power monitoring and protective panels:
 - Open/close control of circuit breakers;
 - Protective functions described on the Drawings;
 - Selection of either remote or local control;
 - Local controls for bus bar disconnectors and circuit breakers; and
 - 8) Provide software with each energy meter without necessity of licensing fees.
- (i) Baseplate
 - (i) Mount Engine Generator Set on a rigid common steel base frame; and
 - (ii) Base frame shall be stiffened to minimise deflections.
 - (j) Vibration Isolators
 - (i) Performance: Limit vibration transmissibility to 10 percent at each frequency.
 - (k) Factory Finishing
 - (i) Engine Generator Set and Instrument Panel: Factory-applied primer and two finish coats of the manufacturer's standard heat-resistant engine paint.
 - (l) Control System
 - (i) Control Panel
 - 1) Rating: IEC 60529 Type IP 23 (minimum);
 - 2) Material: Steel;
 - 3) Instrument Identification: Face label or engraved, black, laminated plastic nameplate with white 6 mm high letters, attached with Type 422 stainless steel screws;
 - 4) Environmentally sealed, solid state, microprocessor based modules for engine control and AC metering; and
 - 5) Provide microprocessor based controller for each generator.

(ii) Instrumentation

- 1) Type: Suitable for engine-mounted vibration environment;
- 2) Mounting: Non-shock mounted;
- 3) Alarm and Signal Contacts: Rated 5 amperes at 230 V AC, dry;
- 4) Vibration Monitoring System; and
- 5) Digital readout for following variables:
 - Engine oil pressure;
 - Engine oil temperature;
 - Coolant temperature;
 - Engine rpm;
 - System DC volts;
 - Engine run hours;
 - Generator AC volts;
 - Generator AC amperes;
 - Generator load in kW;
 - Generator frequency;
 - Number of start attempt;
 - Battery voltage DC volts;
 - Fuel consumption data (total consumed and total consumed since reset); and
 - Percentage load on generator.

(iii) Operator Controls and Indicators

- 1) HANDCRANK/STOP/AUTO/ENGINE TEST selector switch;
- 2) Generator voltage adjustment. (Digital adjustment only. Potentiometer adjustment not acceptable.);
- 3) Single-phase voltmeter to indicate all three phase's voltage;
- 4) Single-phase ammeter to indicate all three phase's ammeter;
- 5) Kilowatt metre;
- 6) Fuel PRESSURE indicator;
- 7) RUNNING TIME indicator;
- 8) Cool down timer;
- 9) Emergency stop pushbutton;
- 10) Cycle cranking adjustment; and
- 11) Battery DC voltage.

- (iv) Alarm Indicators and Shutdown With Manual Pushbutton RESET
 - 1) Low oil pressure (alarm);
 - 2) Low oil pressure (shutdown);
 - 3) Oil pressure sender failure (alarm);
 - 4) High jacket water temperature (alarm);
 - 5) High jacket water temperature (shutdown);
 - 6) Engine temperature sender failure (alarm);
 - 7) Low water level (alarm or shutdown selectable by user);
 - 8) Fail to crank (shutdown);
 - 9) Engine over speed (alarm and shutdown);
 - 10) Engine over crank (shutdown).
 - 11) Low-low fuel tank (alarm and shutdown);
 - 12) High-high fuel tank alarm;
 - 13) Weak battery (alarm);
 - 14) High AC voltage (shutdown);
 - 15) Low AC voltage (alarm);
 - 16) Under frequency (shutdown);
 - 17) Over current (shutdown);
 - 18) Earth fault (shutdown);
 - 19) Generator overload (alarm);
 - 20) Emergency stop (alarm); and
 - 21) Low/High DC voltage (alarm).
- (v) Battery Charger and Batteries
 - 1) Similar to that specified for Starting System except sized for control system requirements.
- (vi) External Interfaces
 - 1) Output Contacts Required:
 - Common fault condition;
 - Generator READY; and
 - Generator RUNNING.
 - 2) Output Contacts: Double pole, double throw relay contacts rated 5 amperes at 230 volts;
 - 3) Accept remote dry start contact closure from Automatic Transfer Switch, rated 10A at 32 V DC; and
 - 4) Synchronizing Panel.

- (vii) Functional Requirements
 - 1) Recracking Lockout: When engine fires, starting control shall automatically disconnect cranking control to prevent recracking for a pre-set period of time after engine stop.
 - 2) Overcranking Lockout: Initiate after adjustable cranking cycle of 1 to 60 seconds or six cranking cycles of 10 seconds on and 10 seconds off.
 - 3) Engine Shutdown upon any of the following conditions: All conditions listed above as causing SHUTDOWN.
- (viii) Special Requirements
 - 1) Mount control system battery charger in Control Panel;
 - 2) Mount battery charger instrumentation on face of Control Panel and match generator instrumentation;
 - 3) Power Requirements: 230 volts, single-phase; and
 - 4) All controls and alarms including vibration monitoring system shall be integrated in to SCADA.
- (m) Generator Distribution Switchgear
 - (i) Generator Distribution Switchgear may not be shown in SLD, but shall be provided as part of this package and installed within the DG set enclosure. The Generator Distribution Switchgear shall be provided as per specification G5.33– High Voltage Switchgear when connected to HV system and G5.34– Low Voltage Switchgear when connected to LV system. Low Voltage Switchboard shall be provided for LV generators and shall be located at switchroom as shown on the drawing.
 - (ii) Current Rating, Measurement and Protection: As shown on the Drawings or as required based on DG set rating.
 - (iii) Short-time Current Withstand Rating: 20 kA for 3 seconds at 6.6 kV, and 65 kA for 3 seconds at 400 V.
 - (iv) Circuit breakers shall meet all applicable requirements of Specification G5.
 - (v) Enclosure
 - 1) Rating: IEC 60529 Type IP 42 (minimum); and
 - 2) Floor mounted.
- (n) Generator Grounding System
 - (i) HV generator shall be grounded through either a high-impedance (transformer/resistor) or low-impedance (resistor) grounding system as shown on the drawing. LV generators shall be solidly grounded, as shown on the drawing.
 - (ii) A high impedance grounding system shall limit ground current with a single-phase fault to a maximum current of 10 amperes.
 - (iii) A low impedance grounding system shall limit ground current with a single-phase fault to a maximum current of 100 amperes.

- (iv) All grounding system components shall be mounted in suitable, weather tight, metal enclosures.
- (v) Provide all current transformers required for generator protection system.
- (vi) Current transformer secondary connections shall be wired to shorting terminals in an IP 65 (minimum) terminal box located on the equipment base plate.
- (vii) When generator is operating in parallel with another generator, only one generator neutral can be earthed at a time, to avoid circulating currents/harmonics. Feature shall be provided for interlocking / isolating the generator neutral earthing through an isolator along with a contactor.
- (viii) The neutral switch shall be capable of carrying for 1 second the prospective earth fault current available when all generators are running in parallel. Auxiliary contacts shall be derived to give a remote indication of the position of each contactor, neutral switch and tripped status.
- (ix) The neutral earthing panel construction and components shall be generally in accordance with the requirements specified for other HV or LV switchboards.
- (o) Ladders, Painting, Safety Signage
 - (i) The diesel engine shall be supplied with simple, light and detachable hot-dipped galvanised access ladders at each doorway. The doorway shall be large enough to allow removal of equipment for maintenance or repair.
 - (ii) Guard shall also be provided around the engine flywheel, generator coupling, generator and exciter of each set. All materials used in the manufacture shall be corrosion resistant.
 - (iii) The container shall be painted blue in colour as per existing containerised generator set. Two coats of rust inhibitor primer and two coats of enamel paint shall be applied.
 - (iv) Safety signage such as “No smoking”, “Fire Extinguisher”, etc shall be paste on the containerised generator set. One set of fire extinguisher of the appropriate type shall be installed outside the container. It shall be mounted inside a red waterproof metal box with lock and the key sealed in a breakable glass.
- (p) Installation
 - (i) One set of installation manual for the generator and all its auxiliaries as prepared by the factory must be delivered together with each generator. The manual must be in English. Any special tools required for proper and correct installation of generator and its auxiliaries and such tools if not available in the local market must also be provided as part of package for the generator. The cost of the set of special tools shall be deemed to be included in the contract sum.
 - (ii) The details of the supervisor who will supervise the full time installation of generator and all its auxiliaries shall be submitted to the S.O. for approval well before the delivery of equipment to site. The supervisor shall have not less than 5 years working experience installing similar generators and shall have written approval from the generator manufacturer or its local sole agent to undertake this job.

- (iii) The ground preparation and generator replacement work shall only commence after the new generator set has been successfully tested at the factory.
- (q) DG Set Enclosure
 - (i) General
 - 1) Provide generator set with outdoor weatherproof enclosure, with entire package with minimum IP 55 rating.
 - 2) Package shall comply with requirements of IEC for wiring materials and component spacing.
 - 3) Design total assembly of generator set, enclosure, and subbase fuel tank (when used) to be lifted into place using spreader bars.
 - 4) Housing:
 - Provide ample airflow for generator set operation at rated load in ambient temperature of 40 degrees C; and
 - Doors:
 - Hinged access doors as required to maintain easy access for operating and service functions; and
 - Lockable and include retainers to hold door open during service.
 - 5) Roof: Cambered to prevent rainwater accumulation.
 - 6) Openings: Screened to limit access of rodents into enclosure.
 - 7) Electrical power and control interconnections shall be made within perimeter of enclosure.
 - 8) Enclosure earthing shall comply in accordance with CP 5 and SS 551.
 - 9) Finishes:
 - Prime sheet metal for corrosion protection and finish painted with manufacturer's standard colour using a two-step electro coating paint process, or equal meeting performance requirements specified below;
 - Prime and paint surfaces of metal parts. Painting process shall result in coating that meets the following requirements:
 - Primer: 0.5 mil to 2.0 mils thick;
 - Top Coat: 0.8 mil to 1.2 mils thick;
 - Gloss:
 - Per ASTM D523, 80 percent plus or minus 5 percent; and
 - Gloss retention after 1 year shall exceed 50 percent.
 - Crosshatch Adhesion: Per ASTM D3359, 4B-5B;
 - Impact Resistance: Per ASTM D2794, 120 inch to 160 inch pounds;
 - Salt Spray: Per ASTM B117, plus 1,000 hours;
 - Humidity: Per ASTM D2247, plus 1,000 hours; and
 - Water Soak: Per ASTM D2247, plus 1,000 hours.

- Painting of hoses, clamps, wiring harnesses, and other non-metallic service parts shall not be acceptable; and
 - Fasteners used shall be corrosion-resistant and designed to minimize marring of painted surface when removed for normal installation or service work.
- 10) Enclosure shall be constructed of minimum 3 mm steel for framework and 2 mm steel for panels.
- 11) Hardware and hinges shall be austenitic stainless steel.
- 12) Exhaust Silencer:
- Install factory-mounted exhaust silencer inside enclosure;
 - Exhaust shall exit enclosure through a rain collar and terminate with a rain cap; and
 - Exhaust connections to generator set shall be through seamless flexible connections.
- 13) Maintenance Provisions:
- Flexible coolant and lubricating oil drain lines that extend to exterior of enclosure, with internal drain valves;
 - External radiator-fill provision; and
 - External fuel fill provision (if equipped with a sub based fuel tank).
- 14) Provide motorized louvers to minimize air flow through enclosure when generator set is not operating. Louvers shall include provisions to prevent accumulation of ice or snow that might prevent operation.
- 15) Inlet ducts shall include rain hoods.
- 16) Necessary stairs shall be provided within enclosure for access to equipment and steps to eliminate tripping hazard.
- 17) Provide external emergency stop switch that is protected from accidental actuation.
- 18) Provide factory mounted and wired electrical distribution panel to serve generator set and enclosure. Provisions required include:
- 100 amp distribution panel board connected to a 120/240V ac utility service;
 - Two duplex GFI receptacles, one inside enclosure, and one weatherproof receptacle on outside of enclosure;
 - Two three-way switches controlling three ac lamps mounted in vapour tight and gasketed fixtures; and
 - Factory-wired normal AC service from panel board to engine coolant and alternator heaters, and battery charger.
- 19) All wiring within the enclosure shall be properly dressed up on cable trays or conduits with marking / tagging of services.

20) Sound Attenuation:

- Provide with sound-attenuated housing which allows generator set to operate at full rated load in an ambient temperature of up to 40 degrees C;
- Enclosure shall reduce sound level of generator set while operating at full rated load to a maximum of 85 dB(A) at any location 1 meters from generator set in a free field environment when tested in accordance with IEC;
- The noise level shall not exceed 85 dB(A) at a hemispherical distance of 3 metres from the exhaust outlet; and
- Insulate enclosure with hydrophilic materials.

(r) Source Quality Control

(i) General

- 1) Conform to SS, IEC, or BS. Factory Acceptance Test shall be carried out in the factory for all routine tests in accordance with IEC standard and witnessed by S.O.; and
- 2) Testing to be in accordance with established ISO 9001 procedures.

(ii) Factory Inspections

- 1) Inspect for required construction, electrical connection, and intended function. All Source Quality Control and Factory Acceptance Tests (FAT) shall be witnessed by S.O.; and
- 2) Notify S.O. at least 2 weeks prior to any testing to allow S.O. to witness tests.

(iii) Factory Acceptance Test

- 1) Routine tests shall be conducted on the generator at the factory of manufacture in accordance with the latest BS or IEC standards. The factory test shall preferably be conducted in Singapore. If S.O. decides to send officer(s) to witness any of the factory tests, the cost of sending the officer(s) including food and lodging (for overseas factory tests) will be borne by the S.O.. All other incidental cost likely to be incurred by the manufacturer shall be deemed to be included in the contract. Details of test procedures and recording format shall be submitted to the S.O. at least 6 weeks before date of tests. The format of the log sheets shall be approved by the S.O.. All measuring instrument to be used for the tests shall be calibrated by an acceptable 3rd party testing organisation not less than 12 months before date of factory tests. Such calibration certificates shall be submitted to the S.O. at least 4 weeks before date of factory tests.
- 2) Whether the S.O. send any officer, its representative or 3rd party to witness any factory test, the results of all factory tests shall be submitted to the S.O. immediately after the completion of tests. Factory tests are preferred to be conducted locally at manufacturer's authorized agent workshop or factory where feasible.

- 3) Each generator unit shall be subjected to the following tests at the manufacturer's works to demonstrate compliance with the relevant Sections of IEC and this Specification. All fuel, lubricants, water, etc shall be supplied and borne by the Contractor and the costs are deemed to be included in the contract sum.

- 25% rating of DG set - 30 minutes
- 50% rating of DG set - 30 minutes
- 75% rating of DG set - 30 minutes
- 100% rating of DG set - 60 minutes
- 110% rating of DG set - 30 minutes

The above test conducted at rated power factor of 0.8:

- Output voltage checks at the rating of DG set;
 - Load rejection of 100% load at rated power factor;
 - Load acceptance to simulate the most demanding site operating conditions stipulated in the specifications and subject to approval of the S.O.. Voltage dip shall be less than 20% and recovery to within 3% of rated voltage not longer than 2 seconds;
 - Operation of all protective devices shall be demonstrated; and
 - Other tests to demonstrate its ability in meeting the performance requirements described in the specifications.
- 4) Contractor shall demonstrate the full compliance with factory requirements at the installation; such as certain critical clearance, levelling and tolerances before the first starting up of engine and energising of alternator.
- 5) All tests shall be conducted and supervised by competent persons complying fully with local regulations, codes of practice and manufacturers' guidelines.
- 6) All protection sensors and operation/protection interlocks shall be demonstrated to be working correctly before the engine and alternator are allowed to be put on load.
- 7) Atmospheric pressure, room temperature and humidity shall be recorded before and after load test. During the load test, the following items and any other items which the Contractor deemed necessary shall be recorded:
- Ampere;
 - Voltage;
 - Frequency;
 - Load/Power;
 - Lubricating oil pressure and temperature;

- Cooling water temperature and flow rate;
 - Engine speed;
 - Fuel consumption and its rate; and
 - Exhaust gas temperature.
- 8) Results shall be measured and recorded at 15-minute intervals.
- 9) Whether or not the S.O.'s representative wishes to witness any factory test, the Contractor must notify the S.O. of all factory tests in writing.
- 10) A high speed recorder shall be used for the measurement of transient conditions. Three copies, including the original of the test results endorsed by the licensed professional engineer responsible shall be forwarded to the S.O. one (1) week after the test.
- (iv) Steady Load Test: Test Engine Generator Set at steady load run of 120 minutes minimum duration at 110 percent full-rated load.
- (v) Transient Load Test: Conduct transient load test to demonstrate ability to meet the load pickup and load release requirements specified.
- (vi) Harmonic Test: Conduct at full load conditions.
- (vii) Record and Report
- 1) Strip chart recording and full harmonic analysis measuring up to the 50th harmonic for both voltage and current and three phases simultaneously;
 - 2) Transient response;
 - 3) Load/speed stability;
 - 4) Engine fuel consumption;
 - 5) Power output; and
 - 6) Harmonic analysis.

G5.41.3 Execution

- (a) Installation
- (i) Level and securely mount Engine Generator Set and the Container or in the building in accordance with the manufacturer's recommendations.
 - (ii) Where applicable, mount Engine Generator Set and the Container on vibration isolators in accordance with isolator manufacturer's recommendations.
 - (iii) The Contractor shall provide additional mitigating measures as may be necessary to assure that generator set vibration does not cause damage to other equipment.
 - (iv) All earthing work, materials, etc shall be carried out in accordance with SS 551 Code of Practice for Earthing.

- (v) Contractor shall install suitable lighting inside every compartments of the containerized generator set. The minimum lighting level shall be of 300 lux. Lightings shall be controlled by one (1) ON/OFF switch located at the entrance of the generator set. The supply for these lights shall be taken from the LV Switchboard.
- (b) Bolts, Nuts Brackets And Supports
 - (i) All bolts, nuts, screws, washers, U-bolts, fasteners, brackets, supports and such fittings used shall be made of stainless steel. The S.O. reserves the right to reject the fittings if the materials are found to be faulty or inferior. The Contractor shall replace the rejected fittings immediately to the satisfaction of the S.O.. All additional costs incurred shall be borne by the Contractor.
- (c) Painting, Colour Code & Label
 - (i) The supplied equipment and plant shall be painted with epoxy paint. The Contractor shall seek approval from the S.O. on the painting system at least one week prior to commencement of the painting works.
 - (ii) The type of colour to be adopted for each plant, equipment and pipeline shall be decided and approved by the S.O..
 - (iii) All the plant, equipment and pipelines shall be labelled. The label on the plant and equipment shall be computer generated stencils with character size and description decided by the S.O..
- (d) Fire Extinguisher
 - (i) One set of fire extinguisher of the appropriate type used in standby generator shall be supplied and stored in an epoxy coated steel storage box with lock. The steel storage box shall be mounted onto the container and above ground. There shall be a transparent cover in front of the box to view the condition of the fire extinguisher. Another small circular transparent cover shall be provided for the key which is mounted inside the storage box.
 - (ii) A PVC signage for fire extinguisher shall be pasted on the container directly above the storage box.
- (e) Instruction Manual And Drawings
 - (i) A total of six (6) sets of operation, servicing and maintenance manuals written in clear and simple English together with drawings bound in stiff cover shall be provided after the successful testing and commissioning of the plants. In addition, one softcopy in CD-ROM format shall be supplied. It shall be inclusive of all the single line drawings.
 - (ii) Each of the manuals shall provide the following minimum information and instructions:
 - 1) Clear, step by step procedures for trouble-shooting, routine servicing, maintenance, repair, dismantling and operation of the system in its entirety. Maintenance/servicing schedules and checklists.
 - 2) A comprehensive lists of parts with part numbers, part names and materials of each equipment.

- 3) Detailed, as-installed drawings of the above equipment. The drawings shall be accompanied with part name and part number.
 - 4) All as-built drawings with wiring diagrams, control schematics, trouble shooting guide and shall include cable routes of all signal and power cables.
 - 5) All test results performed during the commissioning of the system.
- (iii) The instruction manual shall be properly compiled in a hard-cover ring-binder. The sequence of presentation shall be as follows:
- 1) Table of Contents;
 - 2) Original user, maintenance and service manuals from manufacturers of every instrument and equipment;
 - 3) All As-Built drawings and wiring diagrams; and
 - 4) Detail information of after-sales service such as manufacturers' name, contact numbers and addresses for every instrument and equipment, etc.
- (f) The S.O. reserves the right to reject the submission and request for a resubmission if the content is deemed to be too brief, unclear or not precise.
- (g) Field Finishing
- (i) Touch up damaged coating with paint system compatible to existing.
- (h) Field Quality Control
- (i) General: Perform all tests required by SS CP 535, including those identified as quarterly tests.
 - (ii) Site Test:
 - 1) Site tests shall be conducted by the Contractor to verify compliance with all important performance parameters, correct operation and function of the engine, alternator and their auxiliaries and all operation/protection interlocks. This shall include 8 hours reliability test at rated load. The Contractor's commissioning engineer and his staff shall be present throughout the commissioning test and reliability trial runs. The test shall be manned fully by the Contractor and hourly readings shall be taken by the Contractor's staff for all available operation parameters.

- 2) The Contractor shall demonstrate compliance with factory requirements on installation such as certain critical clearance, levelling and tolerances before the very first starting up of engine and energising of alternator. Test and commissioning shall include droop test of the Automatic Voltage Regulator (AVR) and governor to confirm similar settings with the three existing generators. All tests shall be conducted and supervised by competent persons complying fully with local regulations, codes of practice and manufacturers' guidelines. Contractor shall submit detailed test procedures and recording formats for clearance by the S.O. at least 3 weeks before proposed date of test. All protection sensors and operation/protection interlocks shall be demonstrated to be working correctly before the engine and alternator are allowed to be put on load. Contractor shall notify the S.O. of all site tests in writing. The fuel for the site test shall be supplied by the Contractor. The water shall be taken from the source at site. Lubricants shall be borne by the Contractor and shall be topped up to the maximum after the reliability test.

(iii) Free Load Test

- 1) The DG shall be started without load. Any abnormal noise, vibrations, etc shall be checked and rectified. All the running parameters such as lubrication oil pressure, exhaust gas temperature and vibrations, etc shall be checked and readings compared with data obtained during the factory acceptance test. The Contractor shall also check and ensure that all the cylinders are firing and the speed is steady.

(iv) Reliability Test

- 1) Reliability test shall be performed as described below as much as site situation permits.
- 2) The DG shall be gradually put on load, increasing it in steps. As the DG takes on increasing load, all the running parameters shall be checked at each load and readings compared with factory records.
- 3) The DG shall first be tested in solo operation and then in parallel operation with other DGs in the same installation. Only when it is proven that DG can run continuously on $\frac{1}{2}$ full load for 2 hours with all the auxiliaries and controls working properly and stably with other DGs shall then begin the reliability test.
- 4) The reliability test of the DG shall be carried out at $\frac{3}{4}$ to full load for 6 hours (at the S.O.'s discretion) continuous operation to the satisfaction of the S.O.. The DG shall be run in solo or in parallel with the other DGs at site as decided by the S.O.. The reliability test shall start afresh if:
 - The governor settings are adjusted during the test; and
 - Any major interruption happens due to fault developed on the DG.
- 5) A copy of the draft record/data format sheet should be submitted to the S.O. for approval one week before reliability test.
- 6) All operational parameters shall be recorded by the Contractor and submitted to the S.O..

- (v) Auto-Synchronising and Load Sharing Test
 - 1) The new DG shall be operated with each of the existing set for 30 minutes to test its design capability to auto-synchronise and load share with any existing DG (if applicable).
- (vi) Performance Test
 - 1) Perform upon completion of installation;
 - 2) Operate 2 hours minimum at full load at 0.8 power factor lag;
 - 3) Manufacturer's representative shall make necessary adjustments;
 - 4) Demonstrate ability of Engine Generator Set to carry specified loads;
 - 5) Demonstrate Engine Generator Set safety shutdowns; and
 - 6) Additional test requirements per General Specification G5 Electrical Testing.
- (vii) Test Report: Record and report the following:
 - 1) Electric load on generator;
 - 2) Fuel consumption;
 - 3) Exhaust temperature;
 - 4) Ambient air temperature;
 - 5) Safety shutdown performance results; and
 - 6) Sound pressure test.
- (viii) Post-Test Requirements
 - 1) Make final adjustments;
 - 2) Replace fuel and oil filters;
 - 3) Check belt drive tensions; and
 - 4) Demonstrate proper operation of equipment to S.O..
- (i) Manufacturer's Field Services
 - (i) Inspection, Commissioning and Site Adjustment: In addition to the requirements of General Specification G2.25 Manufacturers' Services and General Specification G2.26 Testing and Commissioning, an authorised service representative of the manufacturer shall visit the Site for not less than 5 days per engine generator set to perform the following services:
 - 1) Verify proper installation of the equipment;
 - 2) Inspection, checking and adjusting the equipment; and
 - 3) Commissioning and field testing of the engine generator set.
 - (ii) In addition, an authorised service representative of the switchgear manufacturer shall visit the Site for not less than 3 days to perform the following services:
 - 1) Verify proper installation of the equipment;

- 2) Inspection, checking and adjusting the equipment; and
- 3) Commissioning and field testing of the switchgear.
- (iii) Training of operation Personnel: The authorised service representative or a training instructor shall train the operation personnel in all aspects of generator and switchgear operation and maintenance, including step-by-step troubleshooting procedures with necessary test equipment and in accordance with General Specification G2.27 Training, Advisory and Technical Support Plan.
- (iv) Provide Manufacturer's Certificate of Proper Installation for both the engine generator and the associated switchgear.

END OF SECTION